

Live Demonstration: LDV_UPI system for structural health monitoring of composite material

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Abstract—This live demonstration shows a fully noncontact laser scan system to detect damages of composite material. This system relies on the ability to visualize the laser ultrasonic wave propagation using a high frequency Q-switched laser, a fast laser mirror scanner and a FPGA board for data acquisition and signal processing. For the purpose of the live demonstration, the system is applied to detect damages of a composite leading edge aircraft specimen.

I. INTRODUCTION

One major issue of using composite materials in aircraft structure is the difficulty to inspect the damages made during manufacturing or operating of the aircraft. We introduce in this live demonstration the Ultrasonic Propagation Imager (UPI): a high speed, high resolution, non-contact, non-destructive and intuitive damage inspection system for composite material [1, 2]. A Q-switched laser is guided to scan the structure. The laser ultrasonic wave is detected using a laser Doppler vibrometer (LDV). The signals from the LDV is acquired and process using a FPGA board installed inside a personal computer. A custom-build software controls the operation of the whole process and procedures the results of the inspection.

II. DEMONSTRATION SETUP

The LDV_UPI system depicted in Figure 1 features two parts. Part 1 is a black box includes a LDV sensor head, a Q-switch laser head and two laser mirror scanners placed on a tripod. Part 2 is a control system which contains an amplifier, a LDV controller, a laser controller, and a FPGA board installed inside a personal computer. Part 1 and part 2 are connected with cables.

For the purpose of the live demonstration, the system is applied to detect damages of a composite leading edge aircraft specimen, which placed at the distance of several meters from the tripod. On the composite specimen, several circular retro-reflection tape is stick to its surface to assist the LDV sensing. Those tapes can be easily installed or removed. First, the LDV sensing point is controlled to move to the desired location. Then the inspection area is chosen and the Q-switched laser is guided to scan the structure with a dense grid points. Various parameters such as the speed of the scanning, the frequency bandwidth and the amplification factor can be controlled using a custom user interface. After finish the scanning, the laser ultrasonic wave propagation movie pops out instantly, which allows the user to inspect whether damage exists. During the operation, care should be taken to avoid direct eye-contact with the Q-switch laser.

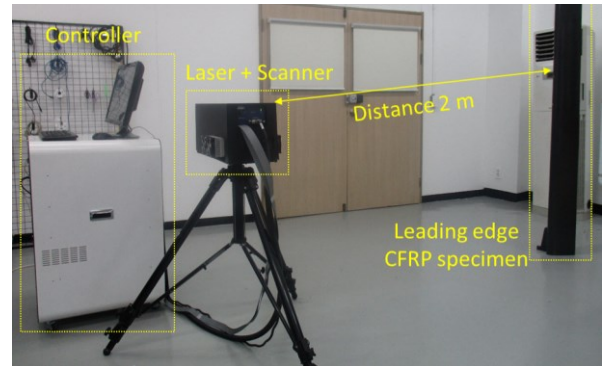


Fig. 1. Set up of the fully noncontact Ultrasonic Propagation Imager (UPI) to inspect damages of a leading edge composite specimen.

III. VISITOR EXPERIENCE

The visitors can interact with the software and experience the damage detection of the composite specimen. The visitors have opportunity of observe the laser ultrasonic wave propagation movie after scanning to see whether the damage was found. The visitors also can change various parameters of the system: scanning rate, frequency bandwidth and amplification factor to understand further the UPI system.

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